

level-4
4

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

(192372301)

I Grdara Teja (Name / Reg No) certify that this submission is my original and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action

Signature of the student: Gr- Teja

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Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve the problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	20 marks
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	20 marks
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	16 marks
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	12 marks
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	8 marks
Presentation	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	4 marks

ASSESSMENT-3

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Smart Home Automation system using Intelligent Systems.

Introduction:

A Smart home automation system controls lighting, temperature and security automatically. It uses sensors such as motion detectors, temperature sensors and time based information to provide comfort, save energy and improve security. Artificial Intelligence uses different types of intelligent agents to make these decisions.

Types of Intelligent Agents

1. Simple reflex Agent

A simple reflex agent acts only on the current input without considering past experiences.

Working in Smart Home:

- Turns on lights when motion is detected
- Turns off lights when no motion is detected
- Starts the AC when temperature exceeds set limit.

Advantages:

- Fast response

- Easy to implement
- low computational cost

Disadvantages

- Cannot learn user habits
- Cannot remember previous states
- may make decisions in complex situations

2. Model-Based Reflex Agent

A model-based agent maintains an internal model of environment and remembers previous states

Working in smart homes:

- remembers whether someone is inside
- Detects if a door is left open
- Adjusts lighting based on room occupancy

Advantages

- works in partially observable environments
- makes better decisions than simple reflex
- uses past information

Disadvantages

- more complex
- requires additional memory.

3. Goal-Based Agent

A Goal-Based Agent selects actions that help achieve a specific goal.

Working in smart-home:

- Keep room temperature at 24°C
- reduce electricity usage.
- ensure home security.

Advantages

- intelligent decision making
- flexible and adaptable
- can plan future actions

Disadvantages

- requires more computation
- slower than reflex agents

4. Utility-Based Agent

A utility based agent selects action that the highest over all benefit (utility)

Working in smart-home:

- Balances comfort and energy savings.
- maintains security while reducing elect consumption.

learns user preferences over time

Advantages:

- Produces optimal decisions
- Handles multiple objectives
- learns user preferences

Disadvantages:

- most complex
- requires more processing power

Best Choice:

A hybrid approach combining model-Based & utility-Based Agents is most effective

Reasons:

- model-Based Agent remembers previous
- utility-Based Agent balances comfort, energy and security
- Together they provide intelligent, adaptive efficient home automation

Conclusion:

A hybrid system ensures maximum reduced energy consumption and better security while learning user preferences over time.

Robot Navigation in an Unknown Maze

Introduction:-

A robot in an unknown maze and must find out with knowing map. Artificial Intelligence uses search algorithms to explore maze and find shortest (or) most suitable path.

Uniform Cost Search (UCS):-

Uniform Cost Search expands the node with lowest path cost first. It always finds the least-cost path.

Working:-

- Starts from initial position
- Expands lowest cost path
- Continues until exit is found

Advantages:-

Finds the optimal path

works with different movement costs.

Complete algorithm

Dis Advantages:-

- uses large memory
- slow for large mazes.

Complexity

Time: Exponential (depends on branching Factor
Path cost)

Space: Exponential.

Iterative Deepening Search (IDS)

IDS repeatedly performs Depth First Search with increasing depth limits until the goal is found

Working:

- Search depth = 0
- Then depth = 1
- Then depth = 2
- Continue until exit is found

Advantages

- low memory usage.
- complete Algorithm.
- good for unknown search depth

Dis Advantages

- repeats node expansion
- Consume more time

complexity

- Time: $O(b^d)$
- Space: $O(b \times d)$

where:

b = branching factor
 d = depth of solution

Intelligent Agents in Maze Navigation

Simple reflex agent:

- Acts only on current surroundings
- cannot remember previous paths.
- may get stuck in loops.

Model-Based Agent

- stores visited locations
- avoids revisiting the same path
- Better exploration

Goal-Based Agent

- plans movements toward the exit
- chooses actions that reach goal.

Utility-Based Agent

- selects safest, shortest, or least expensive

- Balances multiple factors such as time and cost

Best Combination:

- Recommended Agent: goal-based agent with
- Based memory

Recommended Search Algorithm: Uniform Cost Search (UCS)

Justification:

- model based agent remembers explored
- goal-based agent focuses on reaching
- UCS guarantees lowest-cost path
- Together they improve navigation, reduce movement and ensure an optimal

Conclusion:

A goal Based agent with model memory using UCS is best choice for navigation because it remembers explored and find optimal path of exit.